
PHENOVA

MASTER PROGRAM SPECIFICATION

Hardened Competitive Edition • Position A • Independent + Automatic IFEC Enhancement

Core Rule: Build a real, functioning product. Compilation or a rendered UI is never acceptance. Never use fake AI responses, simulated rendering, dummy exports, placeholder buttons, fake persistence, fake authentication, or non-functional screens.

NON-NEGOTIABLE RULES

- PHENOVA must function independently of IFEC (Position A). Core features never depend exclusively on IFEC.
- When registered on IFEC, IFEC automatically becomes an enhancement layer. User does not choose.
- If IFEC is unavailable, PHENOVA falls back to its own AI, backend, local rendering and local projects.
- Every operation must affect real media. No metadata-only editing presented as real editing.
- Edit Plan must be strict, versioned, machine-readable and validated before any media processing.
- Capability Registry is authoritative: a feature may be shown only when the registry declares it available.
- AI Director has hard server-side budgets (operations, media minutes, concurrency, cost).
- All media analysis and AI Director actions respect age tier and content policy.
- No competitive feature is complete merely because a button, screen, API or model call exists.
- Strong production-grade code only. Definition of Done requires real execution verification.

1. Product Identity and Position A

Name: PHENOVA. Android-first, iOS-ready, mobile-first AI-powered video editor with genuine media processing. Architecture is modular and independently testable.

Position A (locked):

- PHENOVA provides its own editing, media processing, projects, storage, authentication, backend, rendering, AI fallback, quotas and subscription logic.
- When registered/connected to IFEC, IFEC automatically becomes an enhancement layer (AI routing, specialist models, heavy rendering, project sync, metering, diagnostics).
- If IFEC is unreachable, PHENOVA automatically falls back to its own capabilities. Core functionality never becomes unusable solely because IFEC is down.
- User-facing state may show “Enhanced by IFEC” or “Running independently” without exposing internal architecture.

2. Architecture

Flutter mobile app; local media engine (FFmpeg or equivalent); AI editing service; backend API; authentication; project service; storage service; rendering/job service; subscription service; automated testing; IFEC integration adapter; Capability Registry; AI Director; canonical project state (timeline + transcript + scene).

Interfaces/adapters allow PHENOVA AI and IFEC AI (and future providers) to be swapped without rewriting the application.

3. Core Editing (Real Media)

- Import multiple videos, photos and audio. Preview source media.
- Trim, cut, split, merge, reorder, delete. Crop, resize, aspect-ratio selection.
- Supported transitions, speed, slow-motion, reverse (document and test exact supported set).
- Text overlays, captions/subtitles, basic colour adjustments.
- Audio volume, mute, mix, dominant-source selection. All audio changes affect actual rendered output.
- Undo/redo in-session required; cross-restart recommended.
- Preview actual edited output. Render and export actual edited video. Every operation must affect real media.

4. Edit Plan (Strict Validation)

Required pipeline: prompt → understand → inspect media → create validated Edit Plan → execute → render → preview → export. Edit Plan is a strict, versioned, machine-readable schema. Every AI-generated plan is validated against the schema, the Capability Registry, entitlements and safety rules before any media processing begins. Invalid, unsafe, impossible or entitlement-violating plans are rejected. AI must cause actual edits, not merely describe them.

5. Capability Registry (Authoritative)

Every editable operation and AI capability is declared in a versioned, machine-readable Capability Registry. Required fields per capability: id, version, maturity (experimental/beta/stable), platforms, resource_cost, requires_network, requires_ifec, entitlement, test_coverage, fallback_behavior. UI and AI Director may expose a capability only when the registry declares it available on the current device, plan and network state. Unsupported analysis or operations must never be presented as fact or available.

6. AI Director / Editing Agent (Hardened)

- AI Director plans and executes multi-step editing tasks from natural-language instructions.
- It must inspect the actual project, media analysis, timeline state, Capability Registry, constraints and user intent before acting.
- It can create a complete first cut, revise it, explain decisions, and apply targeted changes without unnecessarily rebuilding unrelated portions.
- Hard server-side budgets are mandatory: max operations per plan, max media minutes analysed, max concurrent AI Director jobs per user, and cost estimation before execution. Plans exceeding budgets are rejected.
- Maintain an operation graph / edit log: every AI change records affected assets, operations, reason, confidence where available, and rollback relationship.
- Support automatic, guided and manual control modes without creating separate project representations.
- All AI Director actions respect the user's age tier and content policy.

7. Story and Creative Intelligence

- Analyse project-level narrative structure (hook, setup, development, climax, resolution/CTA where applicable).
- Evaluate pacing, repetition, continuity, visual variety, speaker balance, shot quality, audio continuity and unnecessary gaps.
- Creative goals (cinematic, energetic, documentary, educational, promotional, vlog, interview, social short, or user-defined) influence the Edit Plan.
- Provide alternative edit proposals when multiple reasonable interpretations exist (subject to storage and quota limits).
- Distinguish objective corrections from subjective creative choices.

8. Unified Media Understanding

- Index speech/transcript, faces, speakers, objects, scenes, shots, camera motion, silence, audio characteristics, visual quality, metadata and timeline context when supported.
- Support natural-language search (person, object, topic, reaction, location, phrase, visual event).
- Expose only analysis that was actually performed. Cache analysis so expensive processing is not repeated unnecessarily.
- All analysis respects age tier and content policy. Under-18 projects receive stricter analysis limits.

9. Dual Editing Surface (Canonical Project State)

- Provide a conventional timeline and a transcript/document surface over the same underlying canonical project state.
- Edits from either surface remain synchronized. AI actions modify the same canonical representation used by manual editing.
- Single source of truth. Conflict policy: last-write-wins with version history, or explicit user merge when both surfaces edit the same region.
- Users can move from AI automation to frame-level/manual precision at any time.

10. Professional Editing Depth

Expand the Capability Registry over time to include multi-track timeline, keyframes, masks, motion tracking, blend modes, advanced transitions, advanced colour, professional audio processing and compositing where technically supportable. Every new operation must be real, versioned, tested and exposed only when supported by the registry. Add project organisation: bins/folders, tags, labels, markers, searchable metadata and large-project media management.

11. Audio Intelligence

- Speech enhancement, noise/echo cleanup, silence detection, filler removal, room-tone-aware gap handling, volume normalisation, ducking, beat analysis and intelligent music placement where feasible.
- Preserve dialogue intelligibility. Destructive audio decisions require an auditable project operation and, where practical, a preview before commit.
- AI may reason about audio continuity when selecting cuts.

12. Smart Reframing and Multi-Output

- Automatically reframe subjects for multiple aspect ratios while preserving important visual information.
- Generate platform-specific variants from one canonical project without duplicating unrelated editing logic.
- Support user-defined safe areas, subject priority, caption regions and brand layout constraints.
- If the subject cannot be preserved inside the target aspect ratio, the job must fail with a clear error rather than produce a broken or empty frame.

13. Creative Effects and Compositing

Background removal/replacement, object masking, tracking, overlays, controlled restyling, motion graphics and effect stacks as real capabilities when implemented. AI selects effects according to the creative brief. Maintain editable parameters; users can remove or modify AI-selected effects individually.

14. Generative Editing Integration (Provider-Neutral, Future)

- Architect a provider-neutral generative media interface. Do not make generative editing a current dependency.
- Generated assets enter the project as identifiable, editable assets with provenance, provider/model metadata and a visible "AI-generated" flag.
- Original media is never overwritten. Support generative extension/repair only where a compatible provider exists.
- If a generative provider later refuses content, the project must remain consistent and the original timeline intact.

15. Collaboration and Review (Future)

Optional project sharing, comments, review links, approval states and collaborator roles. Preserve immutable project versions. Collaboration inherits the project owner's ownership, quota and age-tier rules. Review notes can be converted into structured AI correction requests.

16. AI Transparency and Control

- AI edit log shows what changed. Accept/reject of proposed changes where the workflow calls for review.
- Targeted undo/restore in addition to global undo. Show capability limitations before execution where possible.
- Never hide an AI failure behind a successful-looking render. Retain provenance for generated content.

17. Reliability, Fallback and Jobs

- Provider architecture allows AI, media analysis, rendering and future generation providers to be swapped without rewriting the project model.
- When a provider fails, retry or fall back according to capability and job policy. A project remains editable if an optional AI provider becomes unavailable.
- Long-running tasks use job states: QUEUED, PROCESSING, COMPLETED, FAILED, CANCELLED. Support progress, cancellation, retry, failure reporting and cleanup.
- Hybrid strategy: short/simple jobs locally; heavy jobs may be backend/IFEC. Device thermal and memory thresholds force offload or graceful degradation.

18. Free Plan and Premium (Server-Enforced)

Free: first month 1 Pro edit/day (1080p first 3 days, then 720p); 2 Basic edits/day from the beginning; after first month 2 Basic/day at 720p; 5 corrections per completed free edit. Basic limits: up to 5 clips, max 5 min per clip, OR one video up to 30 min. Premium: up to 15 clips, max 5 min per clip OR one video up to 2 hours, 1080p, no daily one-edit restriction, unlimited corrections. Exact counting rules for edits, corrections, retries, failed renders and re-renders must be defined and tested. All limits server-enforced.

Pricing: 3-day Premium ■1,500 / \$1; Monthly ■15,000 / \$10; Yearly ■90,000 (USD equivalent configurable). Centralized pricing; never hard-code in client. Verify purchases before granting Premium.

19. Authentication, Storage, Security and Age Tier

- Real email/password authentication, secure hashing, persistent sessions, account recovery, architecture for Google sign-in.
- Every protected resource enforces ownership server-side.
- Age declaration at registration. Under-13 blocked where required. 13–17 receive restricted mode (stricter analysis and content limits). 18+ full mode under normal safety policy.
- Storage abstraction (local + object storage). Validate content, types, sizes, ownership. Prevent path traversal. Per-user and per-project hard quotas. Resumable/chunked uploads. Clean abandoned temporary files.
- Proper release signing. No cleartext HTTP in production. Protect secrets. Rate-limit AI and rendering endpoints. Secure temporary files. Never use debug signing for production.

20. Offline and Degraded Behaviour

Local core editing, preview and previously rendered projects remain accessible offline where technically possible. AI and cloud rendering require network when no local alternative exists. If IFEC is unreachable, PHENOVA falls back to its own capabilities. Network failures must never crash the application. Provide clear degraded-state messaging for every competitive capability that becomes unavailable.

21. Testing Requirements (Expanded)

- Automated tests: authentication, ownership, media, editing, audio, rendering, export, subscription/quota, API, error, retry, persistence, security, IFEC-fallback.
- Real FFmpeg tests required. Do not replace real media tests with metadata-only assertions.
- AI first-cut tests with real media and known structural properties.
- Natural-language correction tests (only requested region changes).
- Transcript/timeline synchronisation tests in both directions.
- Media-intelligence retrieval tests. Pacing/continuity tests. Multi-aspect-ratio reframing tests (subject preservation).
- AI edit-log and reversibility tests. Provider failure tests. Generated-asset provenance tests (when generative is active).
- Large-project performance, thermal, memory and recovery tests.
- Regression tests ensuring new capabilities never violate no-fake-output, quota, security, ownership or Definition-of-Done rules.

22. Implementation Priority

Keep original production foundation phases unchanged. Then: Competitive Phase A — media understanding + canonical project model. Phase B — AI Director / first-cut agent + operation log + budgets. Phase C — transcript/document surface synchronized with timeline. Phase D — professional timeline, audio, colour, compositing, tracking registry expansion. Phase E — smart reframing and multi-output. Phase F — optional collaboration/review. Phase G — provider-neutral generative interfaces when Creative Generation is activated. Each phase obeys Definition of Done: implementation + UI + backend + persistence + error handling + security + tests + real execution verification.

23. Definition of Done and Final Acceptance

A feature is done only when implementation, UI connection, backend integration, persistence, error handling, security, applicable tests and real execution verification are complete. Code existence, screen existence, endpoint existence or successful compilation are never sufficient.

Final acceptance: a real user can install PHENOVA, create an account, import real video/audio, give natural-language instructions, receive a validated executable Edit Plan, perform genuine edits (including AI Director first-cut and corrections), preview, render, export, save and reopen the project, issue further corrections, and export again — both independently of IFEC and with automatic enhancement when IFEC is available. Every visible control performs real work. Quota, age tier, content policy and Capability Registry enforcement are real and server-side.

PHENOVA combines AI agent + transcript + timeline + creative tools + professional controls + media intelligence + reversible history over one canonical project state — while remaining independently functional under Position A.